BRIDGE TO GRADE 6 MATHEMATICS

Student Workbook

Name: ______________________________ Teacher: ______________________________ Dates: ____________________

This workbook is built for summer bridge, tutoring, intervention, or the first weeks of sixth grade. Each lesson includes a warm-up, vocabulary, modeled example, guided practice, independent practice, and an exit ticket.

Readiness Tracker

Part	Skill Target	Before	After	Evidence
1	Place value, decimal operations, estimation			
2	Fraction concepts and operations			
3	Ratios, rates, and percent reasoning			
4	Expressions, equations, and coordinate reasoning			
5	Geometry, measurement, and data			
6	Problem-solving stamina and sixth-grade habits			

Part 1: Number Sense and Decimal Power

Lesson 1: Place Value Flex

Handout

Objective: Read, write, compare, and round whole numbers and decimals using place-value reasoning.

Vocabulary: digit, value, expanded form, decimal, benchmark, round.

Notice: In 48.306, the 3 is worth 0.3 and the 6 is worth 0.006. Explain why.

Model: 7.492 rounded to the nearest tenth is 7.5 because 9 hundredths makes the tenths digit increase.

1. Write 503.079 in expanded form.
2. Compare: 6.205 ___ 6.250.
3. Round 18.946 to the nearest hundredth.
4. Write a number between 4.71 and 4.72.
5. Which is greater: 0.8 or 0.080? Explain.
6. Order: 3.04, 3.4, 3.004, 3.44.

Exit Ticket: Create a decimal with a 7 in the thousandths place and round it to the nearest hundredth.

Lesson 2: Decimal Operations Lab

Handout

Objective: Add, subtract, multiply, and divide decimals while checking answers with estimation.

Model: 4.8 x 0.6 is close to 5 x 0.5 = 2.5, so 2.88 is reasonable.

1. 12.4 + 8.93
2. 30 - 7.86
3. 5.2 x 0.7
4. 18.6 / 3
5. 0.45 x 0.08
6. 9.72 / 0.9
7. A ribbon is 6.4 m long. Four equal pieces are cut. How long is each?
8. Correct the error: 3.5 x 0.2 = 7.0.

Exit Ticket: Solve 24.75 / 5 and write an estimate that proves your answer makes sense.

Lesson 3: Estimation and Error Checks

Handout

Objective: Use rounding, compatible numbers, and inverse operations to find and fix calculation errors.

1. Estimate 198 x 51.
2. Estimate 47.8 / 6.1.
3. Is 6.2 x 9.8 = 607.6 reasonable?
4. Find the missing number: 14.6 + ___ = 22.1.
5. Use inverse operations to check 84 / 7 = 12.
6. Write one mistake students make when lining up decimals.

Reflection: Which checking strategy helps you most?

Part 2: Fraction Confidence

Lesson 4: Equivalent Fractions and Number Lines

Handout

Objective: Generate equivalent fractions and place fractions on number lines.

1. Find three fractions equivalent to 3/4.
2. Place 5/6 between 0 and 1.
3. Which is larger: 7/8 or 5/6?
4. Write 18/24 in simplest form.
5. Explain why 2/3 = 8/12.
6. Create two fractions between 1/2 and 3/4.

Exit Ticket: Draw a number line showing 0, 1/2, 2/3, and 1.

Lesson 5: Fraction Operations Workshop

Handout

Objective: Add and subtract fractions with unlike denominators and explain why common denominators work.

1. 1/3 + 1/6
2. 5/8 - 1/4
3. 2 1/5 + 3/10
4. 4 2/3 - 1 5/6
5. A recipe uses 3/4 cup oats and 2/3 cup flour. How much dry mix?
6. Find and fix: 1/2 + 1/3 = 2/5.

Exit Ticket: Explain one way to find 3/5 + 1/4.

Lesson 6: Multiplication and Division Meaning

Handout

Objective: Connect fraction multiplication and division to area models, groups, and sharing.

1. 2/3 x 3/5
2. 4 x 3/8
3. 1/2 of 18
4. 3/4 / 3
5. 2 / 1/5
6. A trail is 3/5 mile. You walk 2/3 of it. How far did you walk?

Exit Ticket: Write a story problem for 6 / 1/2.

Part 3: Ratios, Rates, and Percent

Lesson 7: Ratio Tables

Handout

Objective: Use ratio language, tables, and diagrams to describe relationships.

1. Write the ratio of 8 red tiles to 12 blue tiles in simplest form.
2. Complete: 3 notebooks cost $6, so 5 cost ___.
3. For every 2 cups rice, use 5 cups water. Complete a table for 1, 2, 4, and 6 batches.
4. Is 4:10 equivalent to 6:15?
5. Draw a tape diagram for 7 girls to 5 boys.
6. Create a ratio that equals 9:12.

Exit Ticket: Explain the difference between part-to-part and part-to-whole ratios.

Lesson 8: Unit Rates in the Wild

Handout

Objective: Find unit rates and use them to compare options.

1. 120 miles in 3 hours = ___ miles per hour.
2. $9 for 6 pens = $___ per pen.
3. Which is cheaper: 8 oz for $2.40 or 12 oz for $3.00?
4. A printer makes 45 pages in 5 minutes. How many in 12 minutes?
5. Write a unit-rate question about sports, music, or shopping.
6. Solve your question.

Exit Ticket: What does "per" mean in a unit rate?

Lesson 9: Percent Benchmarks

Handout

Objective: Use 1%, 10%, 25%, 50%, and 100% to estimate and find percents.

1. 10% of 80
2. 25% of 64
3. 50% of 138
4. 1% of 350
5. Find 15% of 60 using 10% and 5%.
6. A $40 item is 25% off. What is the sale price?

Exit Ticket: Show two ways to find 20% of 45.

Part 4: Expressions, Equations, and Coordinate Thinking

Lesson 10: Expression Builder

Handout

Objective: Translate words into expressions and evaluate them using order of operations.

1. Write an expression: 7 more than a number n.
2. Write an expression: 4 times the sum of x and 3.
3. Evaluate 6 + 3(8 - 5).
4. Evaluate 2a + 5 when a = 9.
5. Write a story for 12 - 3p.
6. Correct: 4 + 2 x 5 = 30.

Exit Ticket: Write and evaluate an expression with parentheses.

Lesson 11: Equation Balance

Handout

Objective: Solve one-step equations and explain each move using balance reasoning.

1. x + 8 = 21
2. p - 9 = 14
3. 5m = 45
4. n / 6 = 7
5. Write an equation for: a number plus 12 is 31.
6. Check your solution to 4y = 32.

Exit Ticket: Why do inverse operations solve equations?

Lesson 12: Coordinate Plane Missions

Handout

Objective: Plot ordered pairs and use coordinates to describe patterns.

1. Plot A(2, 5), B(2, 1), C(6, 1), D(6, 5).
2. What shape is ABCD?
3. Find the distance from A to B.
4. Complete the rule: x increases by 3, y stays 4.
5. Write three points that lie on y = 2.
6. Describe how to move from (1, 3) to (5, 3).

Exit Ticket: Explain why order matters in an ordered pair.

Part 5: Geometry, Measurement, and Data

Lesson 13: Area Decomposer

Handout

Objective: Find areas of rectangles, triangles, and composite figures.

1. Area of a rectangle: 8 by 5.
2. Area of a triangle: base 10, height 6.
3. Draw a composite figure and split it into rectangles.
4. A room is 12 ft by 9 ft. How many square feet?
5. Find missing side: area 48, width 6.
6. Why is triangle area half of a rectangle?

Exit Ticket: Find the area of a 7 by 4 rectangle cut in half diagonally.

Lesson 14: Volume and Nets

Handout

Objective: Use unit cubes, formulas, and nets to reason about rectangular prisms.

1. Volume of a prism: 5 x 4 x 3.
2. How many cubes fit in a 6 by 2 by 2 box?
3. Draw a net for a rectangular prism.
4. Find surface area of a 2 by 3 by 4 prism.
5. What changes when height doubles?
6. Write a real-world volume problem.

Exit Ticket: Explain the difference between surface area and volume.

Lesson 15: Data Displays and Decisions

Handout

Objective: Read, create, and interpret line plots, bar graphs, and measures of center.

1. Find the mean of 4, 6, 8, 10.
2. Find the median of 3, 9, 2, 7, 7.
3. Create a line plot for: 1/2, 1/2, 3/4, 1, 3/4, 1/4.
4. Which measure best describes data with an outlier?
5. Write one conclusion from your line plot.
6. Create a survey question for the class.

Exit Ticket: How can a graph make a comparison easier?

Part 6: Problem-Solving Studio

Lesson 16: Multi-Step Word Problems

Handout

Objective: Read carefully, choose operations, and show organized work for multi-step problems.

1. A class buys 6 packs of markers with 8 markers each and gives away 13. How many remain?
2. A garden is 15 ft by 8 ft. A 3 ft by 4 ft path is added. What is the total area?
3. Tickets cost $7.50. How much for 4 tickets and a $6 snack?
4. Write the question you need to answer before calculating.
5. Circle information that matters in problem 3.
6. Make a new two-step problem.

Exit Ticket: Name one habit that prevents word-problem mistakes.

Lesson 17: Strategy Choice Board

Handout

Objective: Choose efficient strategies: draw a model, make a table, work backward, estimate, or write an equation.

1. Solve with a table: A pattern starts at 5 and adds 4 each step. What is step 10?
2. Solve by working backward: I think of a number, double it, add 9, and get 31.
3. Solve with a diagram: 3 friends share 2 pizzas equally.
4. Solve with an equation: 4 bags have the same number of marbles; total is 52.
5. Which strategy felt fastest?
6. Which strategy made the answer easiest to explain?

Exit Ticket: Match one problem type to one strategy and explain why.

Lesson 18: Grade 6 Readiness Task

Performance Task

Objective: Use number sense, fractions, ratios, equations, geometry, and explanation skills in one task.

Task: Plan a class celebration for 24 students. You have a $120 budget. Choose snacks, supplies, and one activity. Show calculations, justify choices, and create one graph or table.

Requirement	Your Work
Budget calculations	
Ratio or percent reasoning	
Area, volume, or measurement	
Equation or expression	
Written explanation	

Reflection: What are you most ready for in sixth grade?